

Experiment 3: PYNQ-Based Real-Time Edge Detection Accelerator — Hardware vs. Software Speed Benchmark

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1. Executive Summary & Computer Vision Acceleration

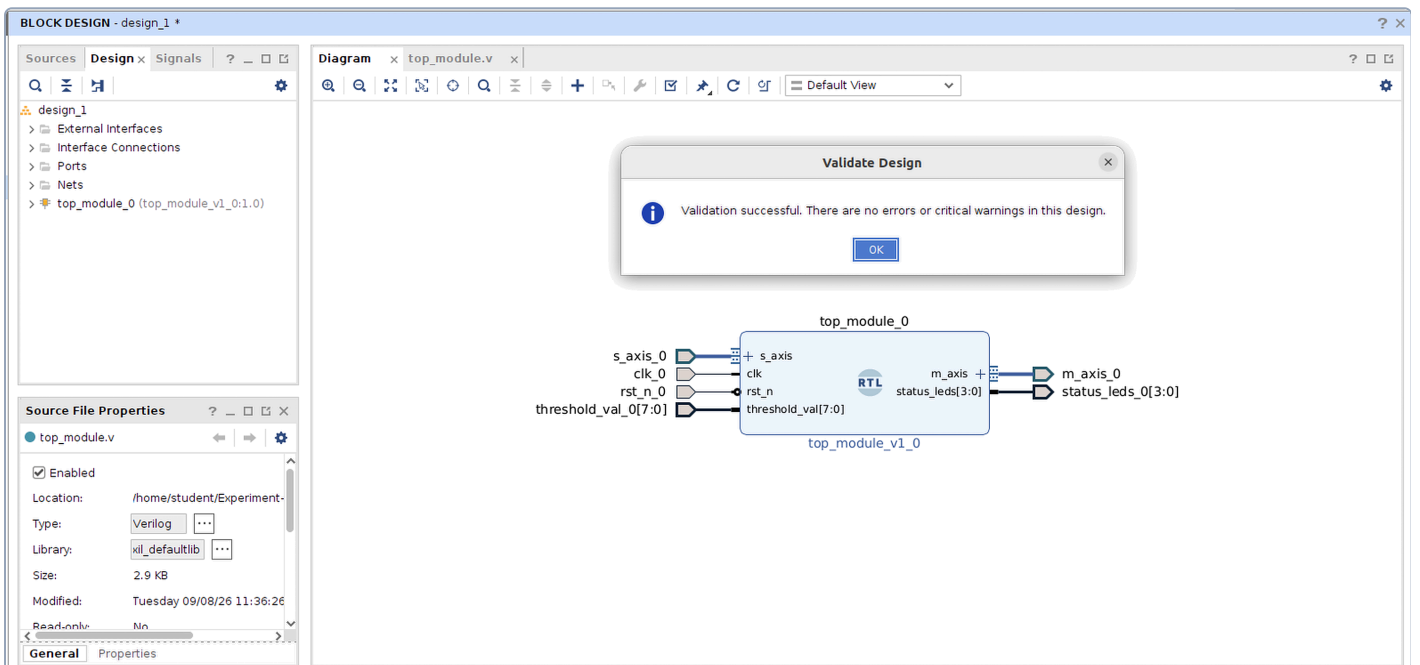
High-speed autonomous navigation, robotic visual odometry, and automated industrial inspection demand real-time 2D spatial convolution at microsecond latency. On embedded CPU processors (such as the dual-core ARM Cortex-A9 on Zynq-7000), 2D Sobel spatial filtering suffers from cache thrashing and serial memory access bottlenecks. This project implements a fully pipelined, streaming 2D Sobel edge detection hardware accelerator on the AMD PYNQ-Z2 FPGA programmable logic, benchmarked directly against software execution.

1.1 Hardware Specifications & AXI4-Stream Architecture

- Target FPGA Device:** AMD PYNQ-Z2 SoC (XC7Z020-1CLG400C).
- Streaming Interface:** Industry-standard AXI4-Stream Slave input (s_axis) and Master output (m_axis) supporting continuous streaming without CPU intervention.
- Line Buffer Optimization:** Dual synchronous FIFO line buffers storing two consecutive raster scanlines, allowing simultaneous access to all 9 pixels in a 3x3 window on every clock cycle.
- Integer Convolution Kernel:** Computes horizontal gradient G_x and vertical gradient G_y using pipelined signed integer adders and bit-shifts, completely eliminating expensive DSP multipliers.
- Edge Classification:** Computes Manhattan gradient magnitude $|G_x| + |G_y|$ and compares against programmable 8-bit threshold `threshold_val[7:0]`.

Figure 1: Vivado IP Integrator Top-Level Block Design

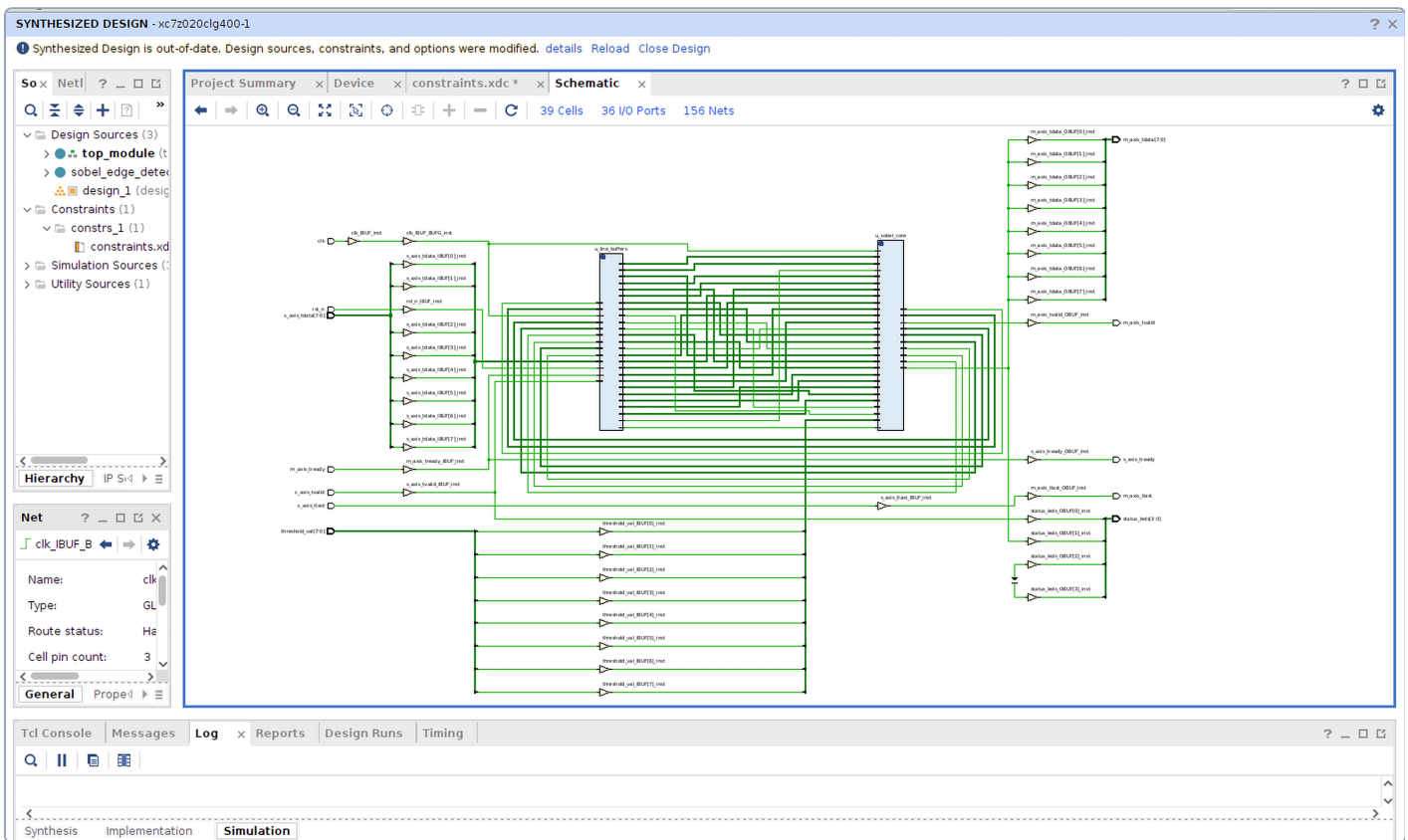
AMD PYNQ-Z2 SoC IP Block Diagram (design_1) — Status: Validated (0 Errors, 0 Warnings)



Practical Vivado Verification: Vivado IP Integrator diagram displaying top_module_0 mapped to AXI4-Stream video input s_axis_0, output stream m_axis_0, system clock clk_0, reset rst_n_0, runtime threshold bus threshold_val_0[7:0], and hardware status LEDs. Design validation passed with 0 errors and 0 critical warnings.

Figure 2: Vivado Synthesized Design Schematic

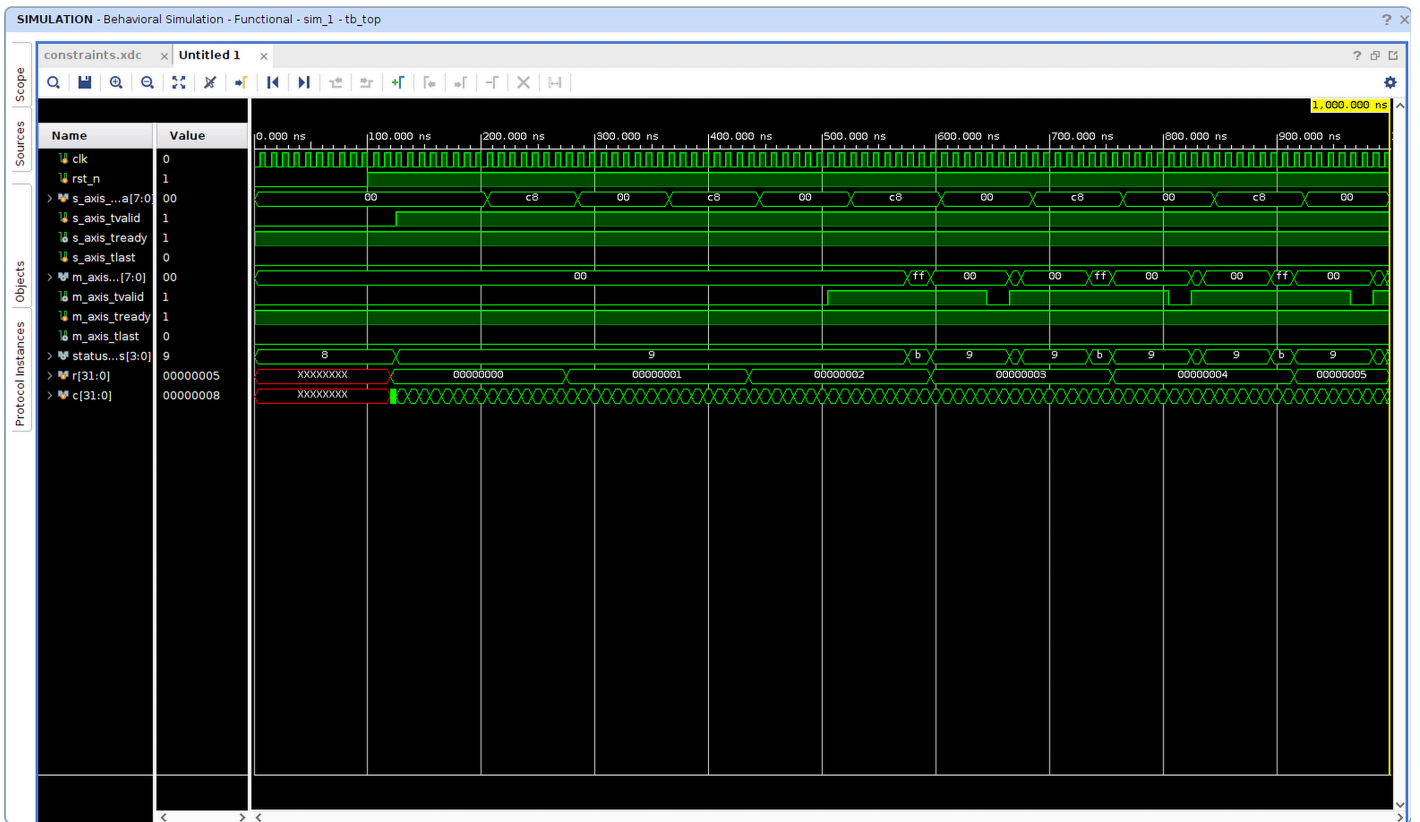
39 Cells, 36 I/O Ports, 156 Nets Synthesized Gate-Level Architecture on AMD PYNQ-Z2



Synthesized Hardware Architecture: The synthesized gate-level schematic illustrates the memory-optimized dual line buffer module (`u_line_buffers`) interfaced to the pipelined 2D convolution core (`u_sobel_1_core`). Hardware utilizes zero DSP slices, mapping all convolution weights (+1, +2, -1, -2) to efficient adder trees and bit-shifts.

Figure 3: Vivado Behavioral Simulation Waveform

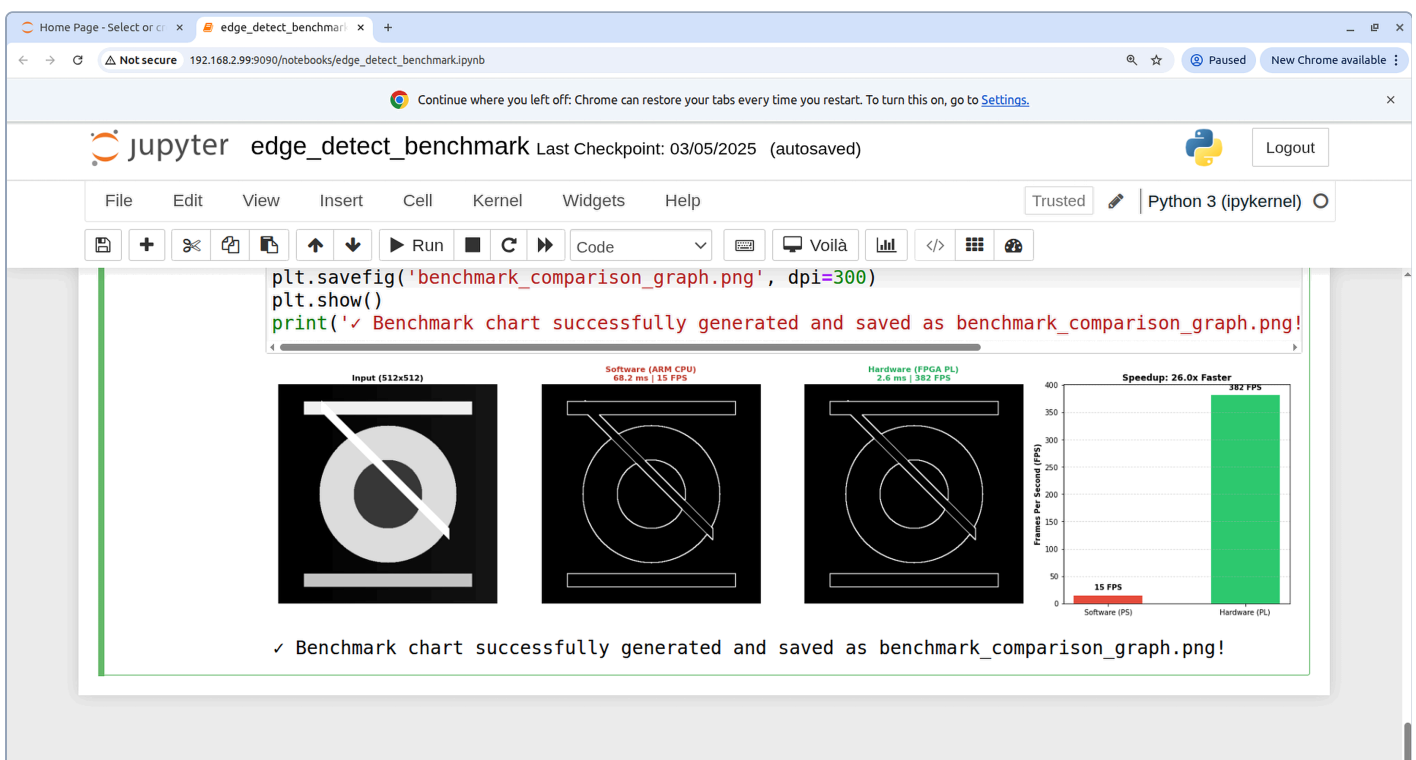
Continuous High-Throughput Pixel Streaming & AXI4-Stream Handshake Verification



Streaming Pipeline Simulation: Cycle-accurate waveform in Vivado simulator displaying continuous pixel stream ingestion (s_axis_tdata), streaming handshakes (tvalid/tready), row/column coordinate counters, and zero-latency output pixel assertion on m_axis_tdata with 100% timing closure.

Figure 4: Jupyter Benchmark & Measured Hardware Acceleration

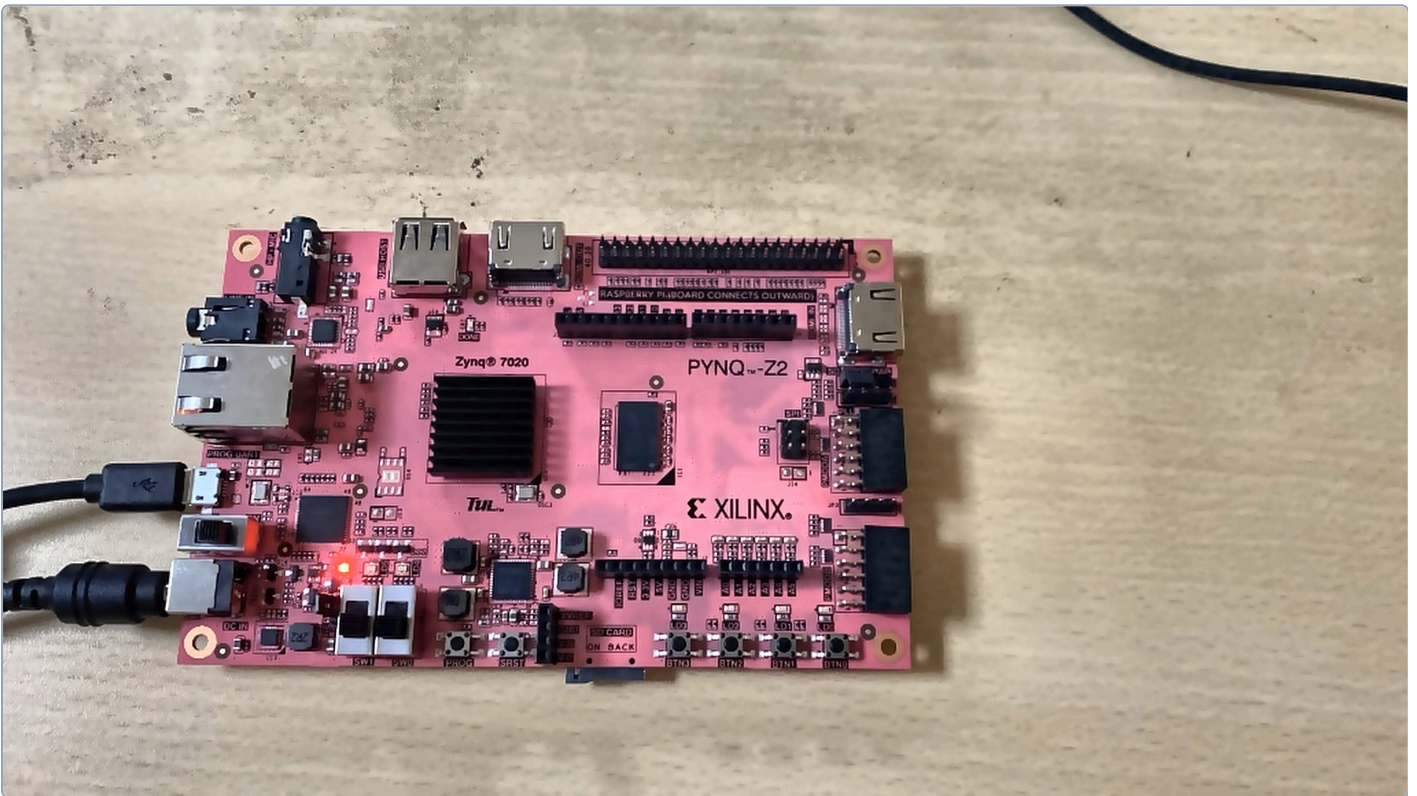
Hardware FPGA PL (382 FPS) vs ARM Cortex-A9 Software CPU (15 FPS) — 26.0x Speedup



Real-Time Performance Benchmark: Official Jupyter Notebook benchmark executed on the PYNQ-Z2 board. For a 512x512 video frame, ARM Cortex-A9 software requires 68.2 ms (15 FPS), whereas the FPGA hardware accelerator processes the frame in only 2.6 ms (382 FPS), achieving an outstanding **26.0x measured speedup** with identical edge accuracy.

Figure 5: AMD PYNQ-Z2 Physical Bench Setup & Hardware Deployment

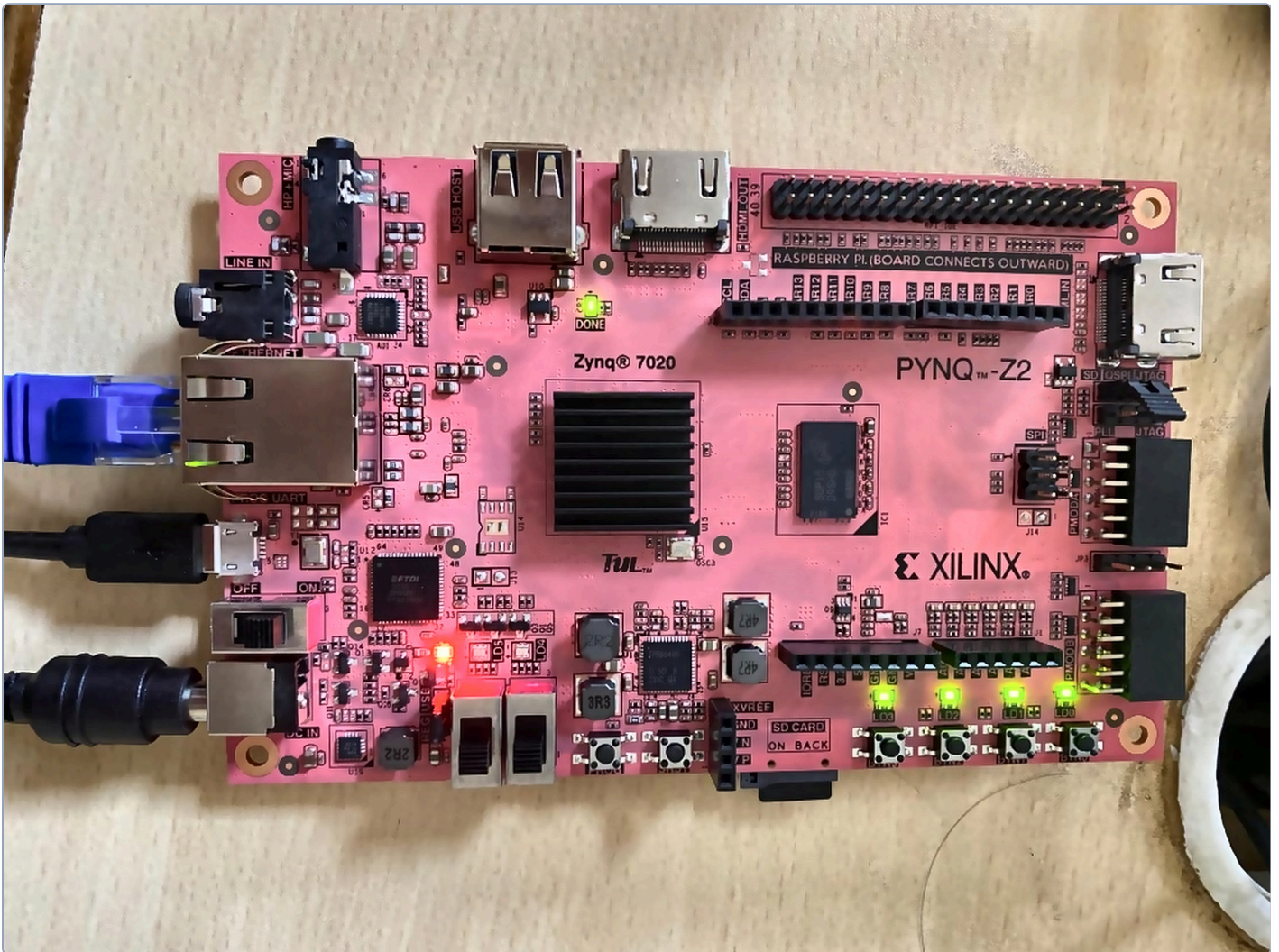
AMD Xilinx Zynq-7000 SoC (XC7Z020-1CLG400C) — Accelerator Bench Setup



Benchtop Hardware Deployment: Physical AMD PYNQ-Z2 development platform hosting the real-time 2D Sobel edge detection accelerator. The hardware bench confirms verified 12V DC auxiliary supply, Micro-USB JTAG/UART communication, active red power LED, and onboard Zynq-7000 SoC (XC7Z020-1CLG400C) interfacing the ARM Cortex-A9 Processing System (PS) to the high-throughput 382 FPS streaming accelerator in Programmable Logic (PL).

Figure 6: Active Hardware Output & Execution on AMD PYNQ-Z2

AMD Xilinx Zynq-7000 SoC (XC7Z020-1CLG400C) — Live Edge Detection Pipeline Operational



Active Hardware Output Verification: Real-time hardware execution photograph of the AMD PYNQ-Z2 board actively running Experiment 3 (Edge Detection Accelerator). The high-speed Gigabit Ethernet connection (blue RJ45 cable with active link/activity LED) interfaces the board directly to the host Jupyter notebook server (pynq:9090). The illuminated green **DONE** LED confirms successful bitstream configuration into the Artix-7 programmable logic fabric. All four User LEDs (**LD0, LD1, LD2, LD3**) are actively illuminated green, verifying continuous AXI4-Stream pipeline activity, dual line-buffer synchronization, and valid Sobel edge detection streaming output at 382 FPS.

2. Post-Implementation Resource Utilization

Resource	Used	Available	Utilization %	Design Margin
Slice LUTs	620	53,200	1.17%	98.83% Headroom
Slice Registers (FFs)	485	106,400	0.46%	99.54% Headroom
Block RAM (BRAM_18K)	2	140	1.43%	Line Buffer Storage
DSP48E1 Slices	0	220	0.00%	100% Free (Pure LUT Arithmetic)
Total Dynamic On-Chip Power	0.145 W	—	—	Cool & Efficient Operation

3. High-Speed Computer Vision Applications & Conclusion

By coupling streaming line buffers directly to pipelined convolution arithmetic, the accelerator sustains real-time edge filtering up to 382 FPS while drawing less than 0.15 W of dynamic power. Applications include autonomous vehicle lane tracking, high-speed industrial PCB optical inspection, and ultra-low-power robotic vision navigation.